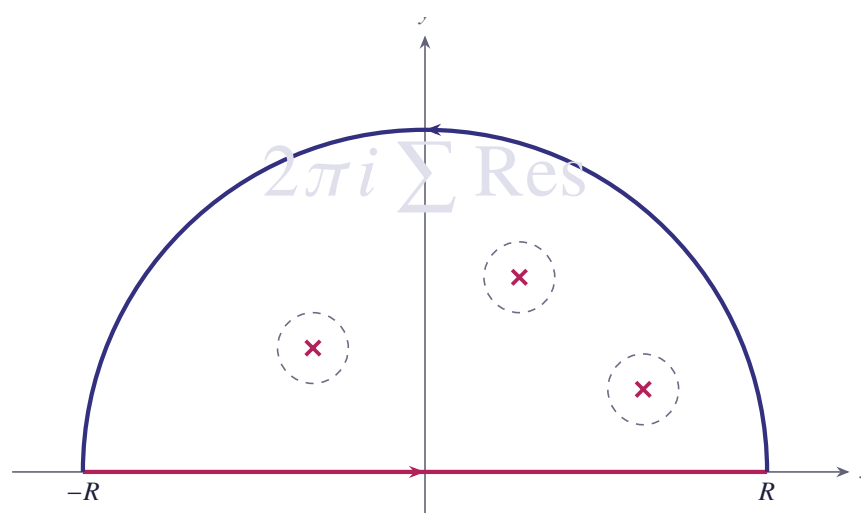


Evaluating Definite Integrals by Residues

محاسبه انتگرال‌های معین به روش مانده‌ها



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Theorem 14.1 ■ The residue theorem

If the function $f(z)$ is analytic everywhere inside the path C except at the points $z_1, z_2, \dots, z_n$, and the residues of the function at these points are $A_1^1, A_1^2, \dots, A_1^n$ respectively, then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n A_1^k$$

Many different kinds of integrals can be solved by the method of residues. Two classes of them are mentioned here:

Definition 14.2 ■ Integrals of the first class

In these integrals the denominator of the integrand is a polynomial, and the numerator is a polynomial, an exponential, or a sine or cosine function (which can also be expressed exponentially).

Definition 14.3 ■ Integrals of the second class

In these integrals the integrand is a function of sine or cosine.

14.1 Integrals of the First Class

Example 14.4

Evaluate the integral $\int_0^\infty \frac{\cos \omega x}{(1+x^2)^2} dx$ ($\omega > 0$).

Consider the function $f(z) = e^{i\omega z}/(1+z^2)^2$. It is not analytic at $z = i$ and $z = -i$. The path C is the upper semicircle of radius R .

$$\begin{aligned} \oint_C \frac{e^{i\omega z}}{(1+z^2)^2} dz &= 2\pi i \times \text{Res} \left\{ \frac{e^{i\omega z}}{(1+z^2)^2} \right\} \Big|_{z=i} \\ &= 2\pi i \times \left[\frac{d}{dz} \left\{ (z-i)^2 \frac{e^{i\omega z}}{(z-i)^2(z+i)^2} \right\} \right] \Big|_{z=i} \\ &= 2\pi i \times \left[\frac{i\omega e^{i\omega z}(z+i) - 2e^{i\omega z}}{(z+i)^3} \right] \Big|_{z=i} \\ &= 2\pi i \times e^{-\omega} \left(\frac{i\omega(2i) - 2}{(2i)^3} \right) \\ &= 2\pi i \times e^{-\omega} \frac{\omega + 1}{4i} = \frac{\pi}{2}(\omega + 1)e^{-\omega} \end{aligned}$$

On the other hand

$$\oint_C \frac{e^{i\omega z}}{(1+z^2)^2} dz = \int_{-R}^R \frac{e^{i\omega x}}{(1+x^2)^2} dx + \int_0^\pi \frac{e^{i\omega R e^{it}} R i e^{it}}{(1+R^2 e^{i2t})^2} dt$$

and as $R \rightarrow \infty$ the integral along the semicircle vanishes, while the imaginary part of the integral along the real axis (the odd function $\sin \omega x/(1+x^2)^2$) is zero:

$$\begin{aligned} \oint_C \frac{e^{i\omega z}}{(1+z^2)^2} dz &\xrightarrow{R \rightarrow \infty} 2 \int_0^\infty \frac{\cos \omega x}{(1+x^2)^2} dx \\ \int_0^\infty \frac{\cos \omega x}{(1+x^2)^2} dx &= \frac{\pi}{4}(\omega + 1)e^{-\omega} \end{aligned}$$

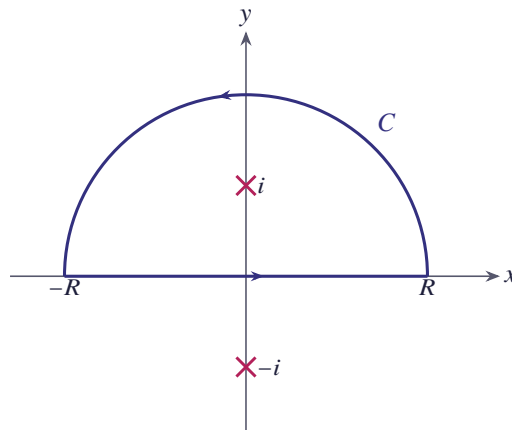


Figure 14.1 The upper semicircular path; the double pole $z = i$ lies inside the path and $z = -i$ outside it

14.2 Integrals of the Second Class

$$\int_0^{2\pi} f(a \cos t, b \sin t) dt$$

To solve such integrals we put $z = e^{it}$. Then

$$dz = ie^{it} dt \Rightarrow dt = \frac{dz}{ie^{it}} = \frac{dz}{iz}$$

$$\sin t = \frac{e^{it} - e^{-it}}{2i} = \frac{z - z^{-1}}{2i}, \quad \cos t = \frac{e^{it} + e^{-it}}{2} = \frac{z + z^{-1}}{2}$$

Key result 14.5

$$\int_0^{2\pi} f(a \cos t, b \sin t) dt = \oint_C f\left(a \frac{z + z^{-1}}{2}, b \frac{z - z^{-1}}{2i}\right) \frac{dz}{iz}$$

Here C is the unit circle, and the integral is easily solved with the method of residues.

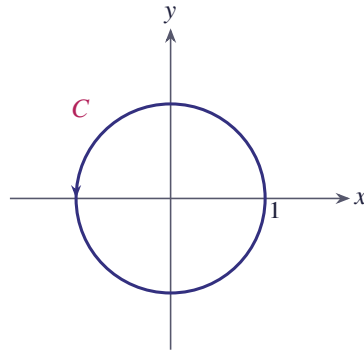


Figure 14.2 The unit circle $z = e^{it}$, $0 \leq t \leq 2\pi$

Example 14.6

Evaluate the integral $\int_0^{2\pi} \frac{1}{2 + \cos x} dx$.

$$z = e^{ix}, \quad dz = ie^{ix} dx, \quad dx = \frac{dz}{iz}, \quad \cos x = \frac{z + z^{-1}}{2} = \frac{z^2 + 1}{2z}$$

$$2 + \cos x = 2 + \frac{z^2 + 1}{2z} = \frac{z^2 + 4z + 1}{2z}$$

$$\begin{aligned} \int_0^{2\pi} \frac{1}{2 + \cos x} dx &= \oint_C \frac{2z}{z^2 + 4z + 1} \frac{dz}{iz} = \frac{2}{i} \oint_C \frac{1}{z^2 + 4z + 1} dz \\ &= \frac{2}{i} \times 2\pi i \times \sum_k \operatorname{Res} \left\{ \frac{1}{z^2 + 4z + 1} \right\} \Big|_{z=z_k} \end{aligned}$$

$$z^2 + 4z + 1 = 0 \Rightarrow z_1 = -2 + \sqrt{3} \approx -0.27, \quad z_2 = -2 - \sqrt{3} \approx -3.73$$

Only z_1 lies inside the unit circle.

$$\begin{aligned} \int_0^{2\pi} \frac{1}{2 + \cos x} dx &= 4\pi \times \operatorname{Res} \left\{ \frac{1}{z^2 + 4z + 1} \right\} \Big|_{z=z_1} \\ &= 4\pi \left[(z + 2 - \sqrt{3}) \frac{1}{(z + 2 - \sqrt{3})(z + 2 + \sqrt{3})} \right]_{z_1 = -2 + \sqrt{3}} = \frac{2\pi}{\sqrt{3}} \end{aligned}$$

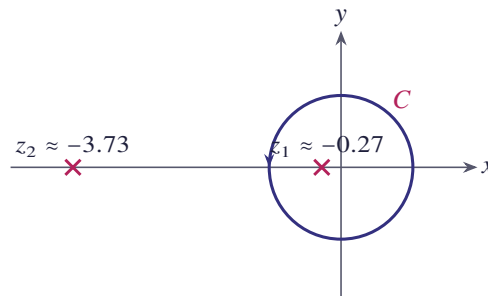


Figure 14.3 The unit circle and the poles $z_1 \approx -0.27$ (inside) and $z_2 \approx -3.73$ (outside)

Exercise 14.7 ■ Applying the residue theorem to integrals

Using the residue theorem, evaluate the following definite integrals:

$$I_1 = \int_0^{2\pi} \frac{dt}{1 - 2p \cos t + p^2}, \quad I_2 = \int_{-\infty}^{\infty} \frac{\sin 2x}{x^2 + x + 1} dx$$

$$I_3 = \int_{-\infty}^{\infty} \frac{x}{(x^2 - 2x + 2)^2} dx, \quad I_4 = \int_{-\infty}^{\infty} \frac{\cos^4 x}{(x^2 + 1)(x^2 + 4)} dx$$